

HEIVA ENGLAND

Health & Economic Inequality Vulnerability Analytics

Professional Data Analytics Project Report

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Professional Data Analytics Project Report

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Role: Data Analyst / Project Developer

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Project proposition: HEIVA England integrates official deprivation, health and economic-participation data through a reproducible analytical pipeline, producing SQL-based analysis, statistical modelling, a composite vulnerability score, geospatial outputs, Power BI reporting and an interactive Streamlit application.

DOCUMENT CONTROL

ITEM	DETAIL
Document	HEIVA England Professional Data Analytics Project Report
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Prepared by	Jirilin Suresh Babu Rajan
Primary audience	Recruiters, hiring managers, data/BI teams, public-sector analytics stakeholders
Analytical scope	England; deprivation, healthy life expectancy, life expectancy, economic inactivity and derived vulnerability analytics
Primary tools	Python, Pandas, NumPy, SQL/SQLite, Statsmodels, scikit-learn, GeoPandas, Power BI, Streamlit, GitHub
Important caveat	The HEIVA Vulnerability Score is an experimental analytical index and is not an official government statistic.

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EXECUTIVE SUMMARY

HEIVA England (Health & Economic Inequality Vulnerability Analytics) is an end-to-end data analytics project designed to examine how deprivation, health outcomes and economic participation overlap across English local authorities. The project was developed as an end-to-end analytical project rather than a single notebook: it combines public-data ingestion, repeatable transformation, relational storage, SQL querying, statistical analysis, historical trends, exploratory forecasting, segmentation, geospatial analysis, business-intelligence reporting, interactive application delivery and automated quality controls.

The central analytical problem is that health, deprivation and labour-market indicators are published in separate systems and at different reporting cadences. HEIVA consolidates these sources using official geographic codes, creating a consistent analytical view that can be used to compare areas and identify places where several forms of disadvantage appear together.

The core output is an experimental HEIVA Vulnerability Score on a 0-100 scale. The score combines normalised deprivation risk, health risk and economic-inactivity risk using equal weights. The score is intentionally transparent and is positioned as an exploratory prioritisation metric, not an official policy index. The analytical layer also includes Pearson correlation, ordinary least squares regression, longitudinal change metrics, a linear trend projection for economic inactivity where sufficient history exists, and K-Means clustering with silhouette-score-based model selection.

The final analytical dataset covers 153 local authorities. Deprivation is strongly negatively correlated with healthy life expectancy ($r = -0.81$) and overall life expectancy ($r = -0.80$). Healthy life expectancy is moderately negatively correlated with economic inactivity ($r = -0.58$). The OLS model accounts for 66.5% of the observed cross-sectional variation in healthy life expectancy ($R^2 = 0.665$; $n = 151$). After adjustment for economic inactivity, deprivation remains statistically significant ($\beta = -0.1005$, $p < 0.001$), whereas the economic-inactivity coefficient does not reach the conventional 5% significance threshold ($\beta = -0.0962$, $p = 0.085$).

Taken together, the results show a clear area-level socioeconomic gradient in health outcomes while also demonstrating that the relationships are not uniform across authorities. HEIVA therefore provides a reproducible framework for identifying areas that warrant closer investigation, rather than a causal model or an official policy classification.

1. BUSINESS CONTEXT AND PROBLEM DEFINITION

Population health and economic participation are closely linked to wider social conditions, but the evidence required to understand those relationships is fragmented. Local-authority analysts, public-health teams, policy researchers and social-impact organisations often need to compare deprivation, healthy life expectancy and labour-market participation across places, yet each indicator may originate from a different source, structure and reporting period.

HEIVA England addresses this information-fragmentation problem by building a reusable analytical layer that standardises the selected indicators at local-authority level. The objective is not to diagnose individuals or claim causality. Instead, the platform provides an evidence-led way to explore whether multiple area-level risks overlap and where deeper investigation may be warranted.

Business Need	HEIVA Response
Fragmented official datasets	Automated ingestion and a unified analytical dataset.
Inconsistent place labels	Joins based on official geographic codes rather than names wherever possible.
Hard-to-interpret multi-dimensional risk	Transparent composite vulnerability scoring and comparative ranking.
Static snapshots	Historical indicator store and change metrics.
Limited stakeholder accessibility	Power BI dashboard and Streamlit application.
Reproducibility concerns	Structured repository, SQL database, QA tests and automated workflow.

Table 1: Business requirements mapped to the HEIVA solution.

2. OBJECTIVES AND ANALYTICAL QUESTIONS

The project was structured around the following delivery objectives:

1. Integrate official public-sector datasets covering deprivation, health and economic participation.
2. Create a reproducible Python ETL pipeline that can be re-run without manual manipulation of intermediate data.
3. Build an analytical SQL layer to support structured querying and downstream reporting.
4. Investigate relationships between deprivation, healthy life expectancy and economic inactivity using exploratory and statistical techniques.
5. Create a transparent composite HEIVA score to summarise overlapping area-level vulnerability.
6. Extend the point-in-time analysis with historical trends, exploratory projections, area segmentation and geospatial analysis.
7. Deliver the analysis through Power BI and Streamlit in a form understandable to non-technical stakeholders.
8. Implement quality controls, tests and workflow automation to improve reproducibility.

2.1. Analytical Questions

ID	Analytical Question	Primary Method
AQ1	Do more deprived areas tend to have lower healthy life expectancy?	Scatter analysis, Pearson correlation, OLS regression
AQ2	Is higher deprivation associated with greater economic inactivity?	Comparative analysis, correlation
AQ3	Are poorer health outcomes associated with lower economic participation?	Correlation, regression, outlier review
AQ4	Which local authorities experience multiple forms of disadvantage simultaneously?	HEIVA score, ranking, geospatial analysis
AQ5	Which areas have similar vulnerability profiles?	Standardisation, K-Means clustering, segment profiling
AQ6	How have selected indicators changed over time?	Historical store, absolute % annualised change

Table 2: Analytical questions and methods.

3. DATA SOURCES AND EXTERNAL CONTEXT

The solution uses authoritative public-sector sources. The English Indices of Deprivation 2025 provides relative deprivation statistics and includes a dedicated upper-tier local-authority summary file. OHID Fingertips provides public-health indicators and an API supporting CSV / JSON retrieval. ONS provides life-expectancy statistics and official local-area context. Geographic boundaries are sourced from the ONS Open Geography Portal.

Source	Role in Project	Key Implementation Note
English Indices of Deprivation 2025 (MHCLG)	Deprivation metrics and upper-tier summaries	Relative measure; interpreted comparatively rather than as an absolute poverty scale.
OHID Fingertips / Public Health Profiles	Healthy life expectancy, life expectancy, economic inactivity and history	API/file ingestion; indicator metadata and time periods retained.
Office for National Statistics	Life-expectancy context and benchmark statistics	Used as an authoritative reference for local/national health context.
ONS Open Geography Portal	Administrative boundary geometry	GeoJSON/boundary joins based on area code.

Table 3: Core data sources.

3.1. External National Benchmark Context:

For context, official current sources report healthy life expectancy at birth in England for 2022-24 of 60.9 years for males and 61.3 years for females. Economic inactivity among people aged 16 - 64 was 21.2% in 2024/25. ONS reports life expectancy at birth in England for 2022-24 of approximately 79.5 years for males and 83.3 years for females.

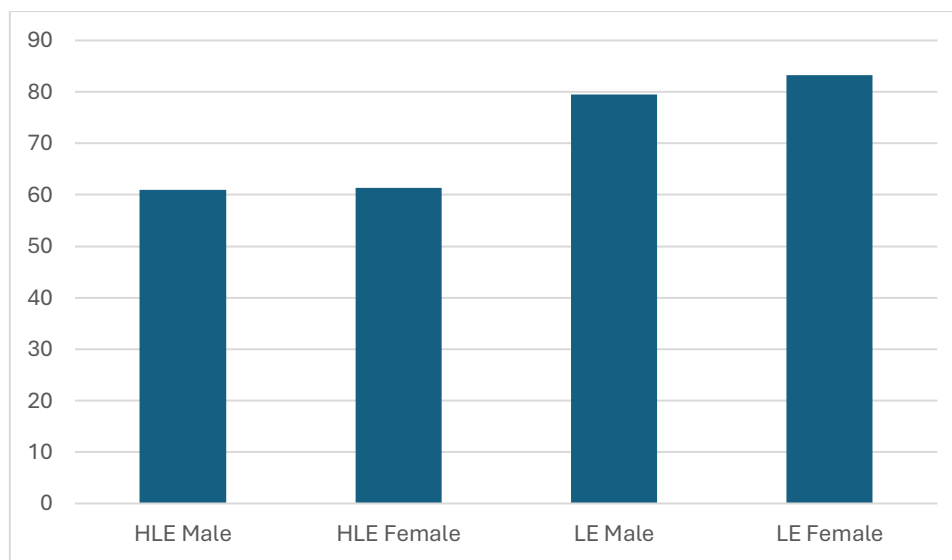


Figure 1: England-level healthy life expectancy and life expectancy benchmarks used for contextual comparison; these values are external benchmarks and are not HEIVA analytical outputs.

4. SOLUTION ARCHITECTURE

The architecture separates source ingestion, transformation, storage, analytics and delivery. This separation is important from an industry perspective because it allows each stage to be validated independently and avoids coupling the dashboard directly to raw source files.

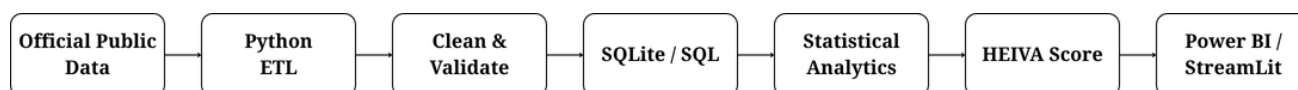


Figure 2: End-to-end HEIVA analytical architecture, from source ingestion and transformation to analytics and stakeholder-facing delivery.

4.1. Architectural Principles

- **Reproducibility:** Raw data are kept separate from processed analytical outputs.
- **Traceability:** Output datasets are derived through named modules rather than manual spreadsheet edits.
- **Stable geography:** Area-code joins are preferred to textual name matching.
- **Separation of concerns:** ETL, statistical analysis, geospatial processing and dashboard delivery are separate layers.
- **Responsible analytics:** Experimental scoring, clustering and forecasting are labelled as analytical tools rather than official classifications.

5. DATA ENGINEERING AND DATA MODEL

5.1. Extraction

The extraction layer downloads the deprivation file and retrieves Fingertips metadata / data. Where possible, indicators are resolved programmatically from metadata rather than permanently hard-coding identifiers. This reduces maintenance effort when source systems evolve and allows the ingestion process to be reused when source metadata changes.

5.2. Transformation and Geography Standardisation

Transformation logic standardises column structures, converts indicator values to numeric form, retains relevant time-period information, selects the appropriate geographic scope and constructs a single-row-per-area analytical snapshot. The key engineering decision is to use official geographic codes for merges. Names are retained for presentation, but codes act as the primary integration key.

5.3. Derived Features

Derived Field	Definition / Purpose
imd_deprivation_percentile	Transforms the deprivation ranking into an intuitive higher-is-more-deprived comparative scale.
healthy_life_expectancy_sex_mean	Persons value where available; otherwise, a pragmatic mean of male/female values for portfolio analysis.
unhealthy_years_estimate	Life expectancy minus healthy life expectancy; an analytical derived measure, not an official indicator.
deprivation_risk	Min-max normalised deprivation component.
health_risk	Reverse-normalised healthy life expectancy component; lower HLE => higher risk.
economic_risk	Min-max normalised economic-inactivity component.
vulnerability_score	Mean of the three risk components multiplied by 100.

Table 4: Key derived analytical fields.

5.4. SQL Data Model

SQLite provides a lightweight relational analytical layer. Core and advanced tables are designed for reusable querying and dashboard support.

Table	Purpose
dim_area	Distinct area-code and area-name dimension.
fact_deprivation	Processed deprivation metrics.
fact_indicator_latest	Long-form latest indicator observations.
analytics_snapshot	Convenient cross-sectional joined view.
analytics_enriched	HEIVA score and segment-enriched analytical table.
indicator_history	Historical observations by area, indicator and period.
area_trends	First/latest values and change metrics.
cluster_profiles	Average analytical profile for each data-driven segment.
economic_inactivity_forecast	Illustrative short-horizon trend projections.

Table 5: Analytical database tables.

6. ANALYTICAL METHODOLOGY

6.1. Exploratory Data Analysis

EDA is used to understand distributions, missingness, ranges and relationships before inferential interpretation. The workflow includes descriptive statistics, highest- and lowest-ranked areas, scatterplots, cross-area comparisons and missing-data checks. This is a critical control because public datasets often contain different reporting periods, sex breakdowns and area-type structures.

6.2. HEIVA Vulnerability Score

The HEIVA score provides a concise summary of three project dimensions: deprivation, healthy-life-expectancy risk and economic-inactivity risk. Each variable is placed on a common 0-1 scale before aggregation. Equal weighting is used because it is transparent and avoids implying an empirically validated priority between the three dimensions.

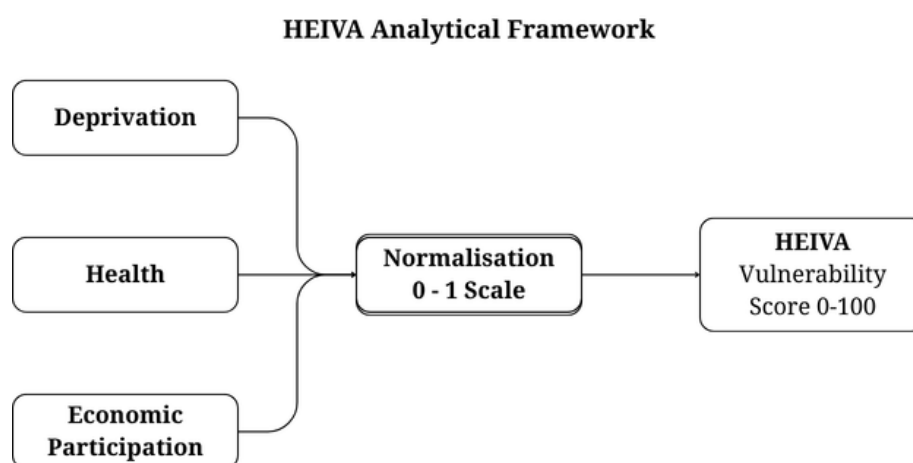


Figure 3: Conceptual construction of the HEIVA Vulnerability Score using normalised deprivation, health and economic-participation risk components.

Interpretation rule:

A high score means that, relative to the other areas in the analytical dataset, an authority has a stronger combination of deprivation, poor healthy life expectancy and economic inactivity. It is an exploratory comparison tool, not a measure of individual-level risk.

6.3. Correlation Analysis

Pearson correlation is used to quantify linear association between selected continuous indicators. It supports the descriptive questions about whether more deprived areas tend to have poorer health outcomes or different economic-participation profiles. Correlation coefficients are reported with direction and magnitude, but they are not interpreted as causal effects.

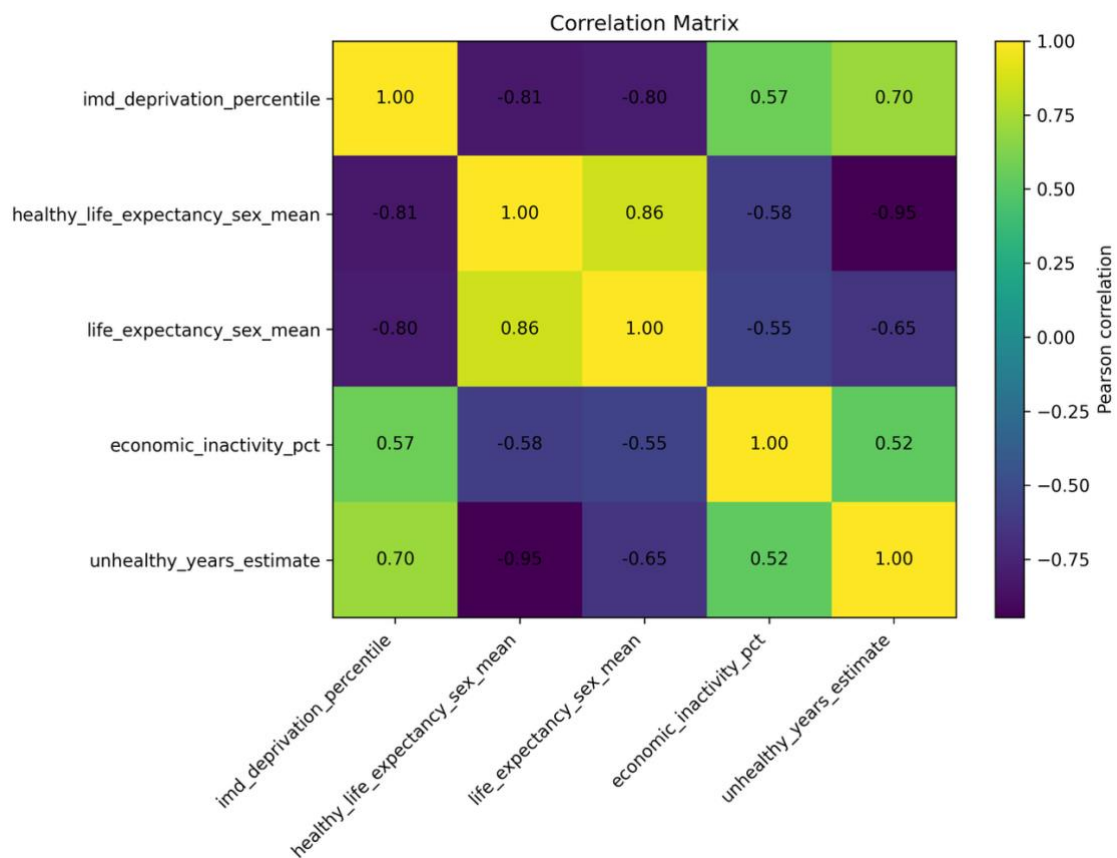


Figure 4: Pearson correlation matrix for the core HEIVA analytical indicators.

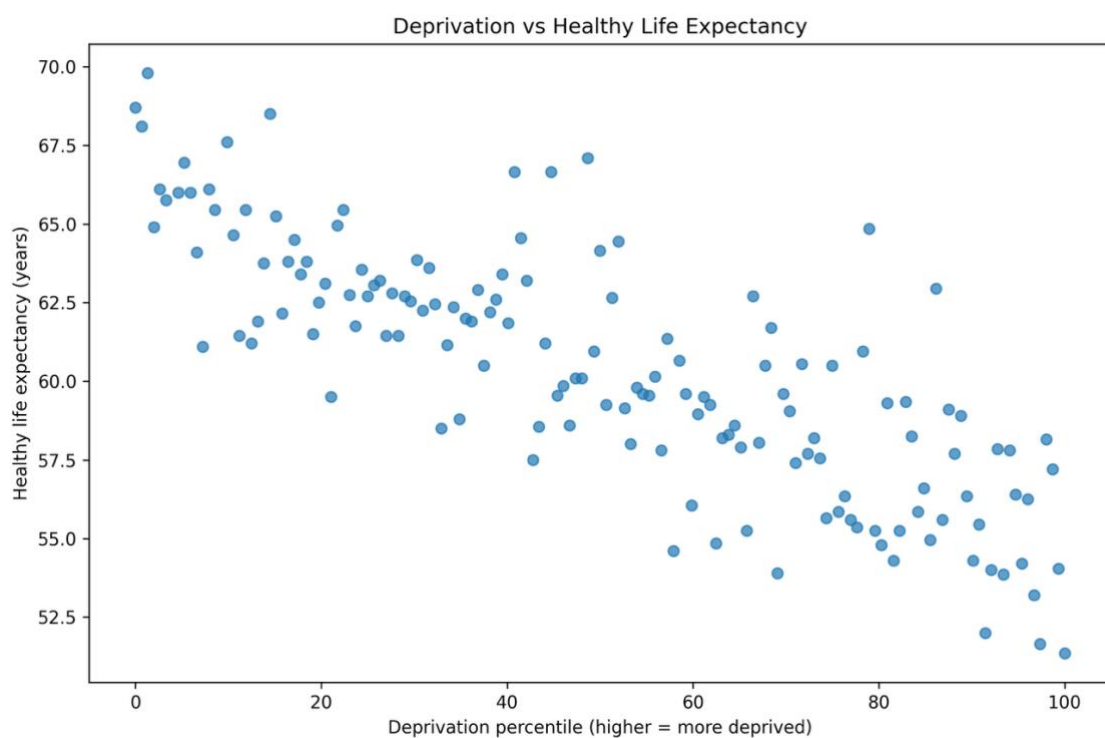


Figure 5: Relationship between the HEIVA-derived deprivation percentile (higher values indicate greater relative deprivation) and healthy life expectancy across English local authorities.

6.4. Regression Analysis

The project applies an ordinary least squares model in which healthy life expectancy is modelled as a function of deprivation and economic inactivity. The model is used to examine conditional association and to produce an interpretable R-squared and coefficient set. Statistical significance, if reported, should be interpreted alongside effect size, model fit, sample size and data limitations.

Model statement:

$$\text{Healthy Life Expectancy} = \beta_0 + \beta_1(\text{Deprivation}) + \beta_2(\text{Economic Inactivity}) + \text{Error}.$$

This is an association model; it does not establish that either explanatory variable causes a change in healthy life expectancy.

7. ADVANCED ANALYTICS

7.1. Historical Trend Layer

The advanced pipeline retains available historical observations rather than using only the latest point. For each area and indicator, it calculates the first and latest observed values, absolute change, percentage change, annualised change and number of observations. This allows the analyst to distinguish high current vulnerability from adverse or improving direction of travel.

7.2. Illustrative Economic-Inactivity Projection

Where sufficient annual history exists, a simple linear regression is fitted to period-end year and economic inactivity. The next one- and two-year values are generated as illustrative trend projections. The method is intentionally transparent; it is not positioned as a macroeconomic forecast because it does not model policy intervention, shocks, demographic change or structural labour-market dynamics.

7.3. Area Segmentation

K-Means clustering groups areas with similar multivariate profiles. Features are standardised before clustering and missing numeric values are median-imputed. Candidate cluster counts are evaluated with silhouette scores rather than being chosen arbitrarily. Cluster labels are then ordered and given interpretable names using average HEIVA score and feature profiles.

7.4. Geospatial Analysis

An official ONS boundary file is joined to the enriched analytical dataset using geographic codes. The output GeoJSON supports interactive choropleth mapping, enabling users to move from a league-table view of vulnerability to a spatial view of regional concentration and neighbouring-area patterns.

8. BI, APPLICATION AND DELIVERY LAYER

The project deliberately delivers the analysis in more than one interface. Power BI demonstrates business-intelligence capability and structured KPI / report design, while Streamlit demonstrates the ability to productise Python analytics for interactive exploration. Both depend on the same validated analytical layer rather than separate manually prepared datasets.

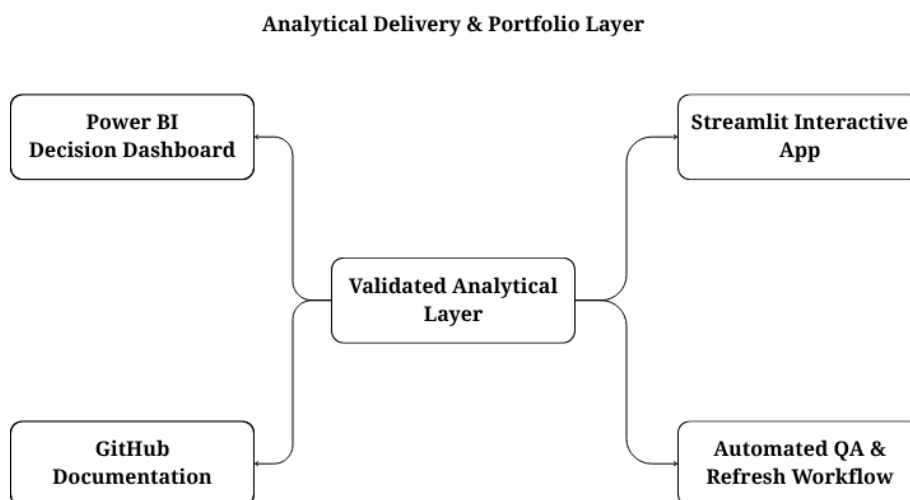


Figure 6: HEIVA analytical delivery layer integrating Power BI, Streamlit, GitHub documentation and automated QA.

8.1. Power BI Dashboard

Page	Purpose	Example Components
England Overview	Executive summary of the analytical landscape	Area count, HLE, inactivity, HEIVA KPIs, ranking, map, scatterplot
Health Inequality	Focus on health outcomes	HLE, life expectancy, derived years outside good health, sex comparison
Economy & Deprivation	Focus on labour market and socioeconomic context	Economic inactivity, deprivation, comparative scatterplots and ranking
Local Authority Explorer	Drill-down for a selected area	Area slicer, HEIVA score, health/economic/deprivation indicators, benchmark comparison

Table 6: Power BI page design.

8.2. Streamlit Application

The Streamlit application presents Overview, Historical Trends, Area Segments, Area Explorer and Methodology pages. The design allows the analytical methodology and caveats to sit alongside the charts rather than being hidden in separate documentation. This allows users to explore results while retaining access to methodological caveats.

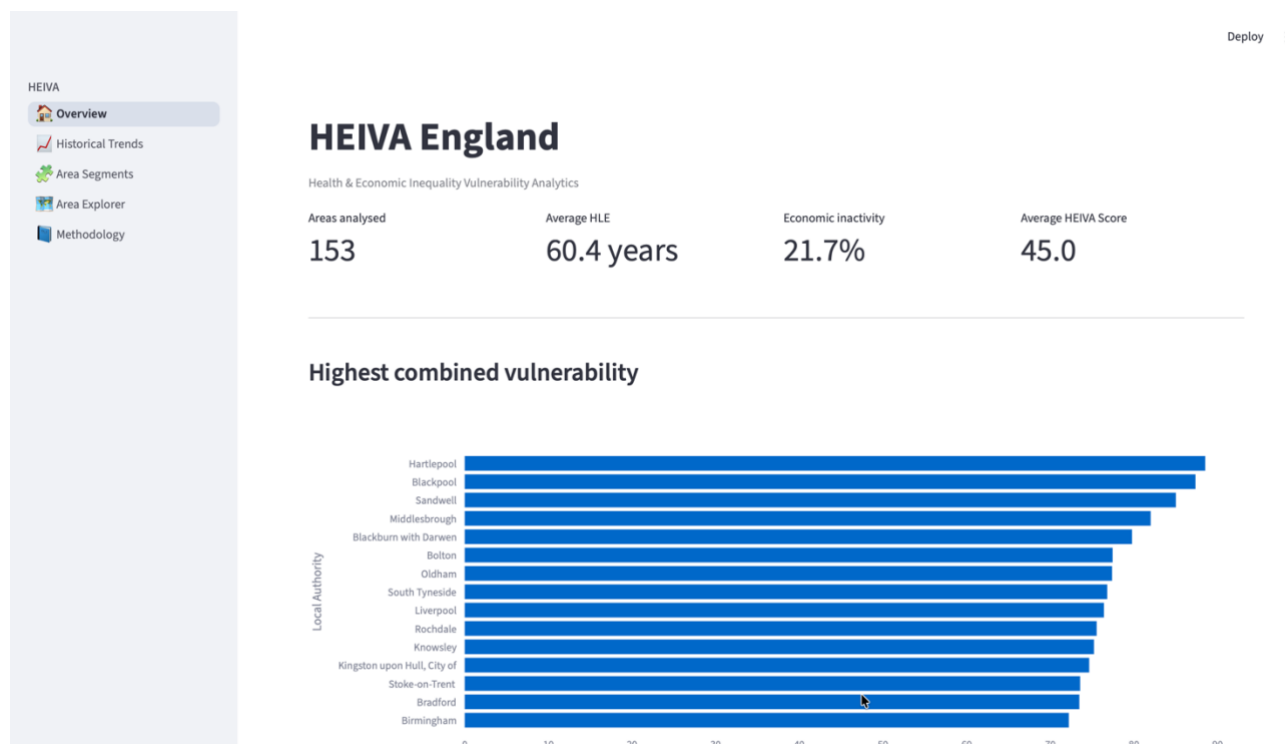


Figure 7: HEIVA England Streamlit application showing headline indicators and the combined-vulnerability ranking.

9. DATA QUALITY, REPRODUCIBILITY AND AUTOMATION

HEIVA applies explicit data-quality controls before analytical outputs are released. Programmatic checks cover duplicate geographic codes, core-indicator missingness, HEIVA score bounds and the availability of historical observations. Automated pytest tests provide a repeatable acceptance gate, while the machine-readable quality report records the resulting QA metrics.

Control	Purpose	Example Acceptance Criterion
Area-code uniqueness	Protect one-row-per-area design	No duplicate area codes in the enriched snapshot.
Core-field completeness	Detect integration failures	Missingness is measured and any material gaps are explained.
Score validation	Prevent composite-score defects	All non-null HEIVA scores lie between 0 and 100.
Historical availability	Protect trend analysis	History exists for the selected indicators before trend / forecast execution.
Automated tests	Repeatable release check	<i>pytest</i> passes before portfolio release.
GitHub Actions refresh	Operational reproducibility	Pipeline can be run manually / scheduled in a clean environment.

Table 7: Data-quality and reproducibility controls.

9.1. Automation

The GitHub Actions workflow demonstrates how the project can be refreshed in a clean environment: checkout, Python setup, dependency installation, pipeline execution and refreshed-output commit where configured. The workflow should be scheduled at a cadence that reflects public-data update frequency rather than at unnecessarily high frequency.

9.2. Verified Data Quality Metrics

QA Measure	Result	Interpretation
Duplicate area codes	0	One-row-per-area requirement satisfied
IMD deprivation missingness	0.00%	Complete deprivation coverage
Healthy life expectancy missingness	1.31%	Very small level of missing data
Economic inactivity missingness	0.65%	Very small level of missing data

Vulnerability-score missingness	0.00%	No null HEIVA scores in the generated output; component-level completeness is assessed separately
HEIVA score range validation	Passed	Scores lie within expected 0-100 bounds
Historical observations	11,095	11,095 historical observations available for longitudinal analysis
Historical indicators	3	Health / economic history retained
Areas in master dataset	153	Final analytical geography coverage

Table 8: Data-quality metrics

Deprivation data achieved complete coverage, while healthy life expectancy and economic inactivity recorded missingness rates of 1.31% and 0.65%, respectively. The generated HEIVA score contained no null values and passed the implemented 0-100 range validation. Component-level missingness is reported separately because score availability alone does not establish completeness of every underlying indicator.

10. RESULTS AND FINDINGS

Metric / Finding	Final Project Value	Where To Retrieve It
Number of local authorities analysed	153	analytics_master.csv / analytics_enriched.csv
Core analytical rows after final join	153 areas	analytics_master.csv
Correlation: deprivation vs healthy life expectancy	$r = -0.81$	outputs/tables/correlation_matrix.csv
Correlation: healthy life expectancy vs economic inactivity	$r = -0.58$	outputs/tables/correlation_matrix.csv
Regression R-squared	$R^2 = 0.665$	outputs/reports/regression_summary.txt
Most vulnerable authority by HEIVA score	Hartlepool - 88.50	outputs/tables/vulnerability_ranking.csv
Lowest-vulnerability authority by HEIVA score	Wokingham - 6.77	vulnerability_ranking.csv
Selected K-Means cluster count	2	outputs/tables/clustering_metadata.json
Best silhouette score	0.4561	clustering_metadata.json

Table 9: Summary of verified analytical results.

10.1. Key Analytical Findings

Finding 1 - Deprivation shows a strong association with healthy life expectancy:

The strongest core relationship identified in the cross-sectional analysis was between deprivation percentile and healthy life expectancy ($r = -0.81$). More deprived local authorities generally recorded lower healthy life expectancy. The corresponding scatterplot demonstrates a pronounced downward trend, although the remaining spread illustrates meaningful variation between areas.

Finding 2 - Deprivation is also strongly associated with overall life expectancy:

The correlation between deprivation and overall life expectancy was $r = -0.80$, indicating that the observed socioeconomic gradient extends beyond healthy life expectancy to overall longevity.

Finding 3 - Health and economic participation are related, but the relationship is more moderate:

Healthy life expectancy and economic inactivity were negatively correlated ($r = -0.58$), while deprivation and economic inactivity were positively correlated ($r = 0.57$). These patterns indicate that poorer health, greater deprivation and lower labour-market participation tend to overlap at area level.

Finding 4 - Deprivation retains the stronger conditional association in the OLS model:

The model was estimated using 151 observations and achieved $R^2 = 0.665$ (adjusted $R^2 = 0.661$). After accounting for economic inactivity, deprivation remained negatively associated with healthy life expectancy ($\beta = -0.1005$, $p < 0.001$). Economic inactivity also had a negative coefficient ($\beta = -0.0962$), but the estimate did not reach the conventional 5% significance threshold ($p = 0.085$). The model therefore provides stronger evidence for an association between deprivation and healthy life expectancy than for an additional economic-inactivity association after deprivation is included.

Finding 5 - Several authorities exhibit high combined vulnerability:

Hartlepool recorded the highest HEIVA score (88.50), followed by Blackpool, Sandwell, Middlesbrough and Blackburn with Darwen among the highest-ranked authorities. At the opposite end of the distribution, Wokingham recorded the lowest score (6.77). The ranking identifies areas where deprivation, health and economic-participation risks overlap most strongly within the HEIVA framework; it should not be interpreted as an official classification of local-authority performance. HEIVA scores ranged from 6.77 to 88.50, with a mean of 45.02 and a median of 43.91. The interquartile range was 29.37-61.63, demonstrating substantial variation in the combined vulnerability measure across the analysed authorities.

Finding 6 - Quality checks identified low missingness and no duplicate geographic codes:

The final master dataset contains 153 local authorities and no duplicate geographic codes. Missingness was 1.31% for healthy life expectancy and 0.65% for economic inactivity, while deprivation data had complete coverage. The historical layer contains 11,095 observations across three indicators.

Finding 7 - A two-cluster solution provided the clearest segmentation among the tested configurations:

K-Means models with two to five clusters were evaluated using silhouette scores. The two-cluster configuration achieved the highest score (0.4561) and was therefore retained. The score indicates moderate rather than strong cluster separation, so the segments are used as exploratory area profiles rather than definitive classifications.

11. BUSINESS AND SOCIAL VALUE

HEIVA is most useful as a prioritisation and communication layer. It does not allocate resources automatically; instead, it allows an analyst or stakeholder to identify areas where health and socioeconomic indicators align in potentially concerning ways and then investigate those areas with additional evidence.

Stakeholder	Potential Use
Local authority / public-health analyst	Benchmark areas, investigate inequality, identify outliers and communicate findings.
NHS / integrated-care analytics team	Use wider determinants as context for population-health planning.
Policy / social-impact organisation	Target deeper qualitative or quantitative investigation toward areas with overlapping disadvantage.
Consultancy / BI team	Demonstrate a repeatable approach to integrating public data into decision-support products.
Recruiter / hiring manager	Evaluate practical Python, SQL, BI, statistical, geospatial and communication skills in one project.

Table 10: Stakeholder value.

12. GOVERNANCE, ETHICS AND LIMITATIONS

12.1. Correlation and Causality

Observed area-level relationships are associative. The project does not establish that deprivation or economic inactivity causes a specific health outcome.

12.2. Ecological Fallacy

The unit of analysis is a geographic area. A high-vulnerability authority contains diverse residents; the score must not be applied to individuals or treated as an individual risk classifier.

12.3. Mixed Reporting Periods

The latest available indicators may refer to different reporting periods. The cross-sectional layer should therefore be described as a latest-available-observation analysis, not a perfectly contemporaneous measurement.

12.4. Relative Deprivation

The English Indices of Deprivation measure relative deprivation. Ranks or derived percentiles are comparative and do not provide a cardinal measure of absolute hardship.

12.5. Composite-Score Assumptions

Min-max normalisation and equal weighting are modelling choices. The score is transparent but not externally validated. Sensitivity analysis would be required before any policy-grade use.

12.6. Forecasting and Clustering

Trend projection assumes continuation of historical direction and does not model structural shocks. Clustering depends on variables, preprocessing and algorithm choice. Both are exploratory.

12.7. Data Privacy and Licensing

The project is based on aggregate public data rather than personally identifiable records. Source-data licensing and attribution conditions must still be respected, including relevant Open Government Licence terms.

13. IMPLEMENTATION AND OPERATING MODEL

13.1. Standard Run Sequence

1. Create / activate the Python virtual environment and install requirements.
2. Place the official ONS geography boundary file in *data/geospatial/utla_boundaries.geojson*.
3. Run python *main.py* to refresh extraction, transformation, analytics, QA and database tables.
4. Run *pytest -v* and resolve any failed quality gates.
5. Launch *streamlit run app.py* for interactive review.
6. Refresh Power BI from the final processed analytical dataset / database as configured.
7. Capture final dashboard screenshots and update the README and report results table.
8. Commit and tag the final release in GitHub.

13.2. Primary Project Artefacts

Artefact	Operational Role
analytics_master.csv	Core cross-sectional dataset.
analytics_enriched.csv	HEIVA score and segment-enriched dataset.
health_economic_inequality.db	SQL analytical database.
indicator_history.csv	Historical indicator observations.
area_trends.csv	Change metrics.
economic_inactivity_forecast.csv	Illustrative projection output.
cluster_profiles.csv	Segment summaries.
heiva_map.geojson	Geographic analysis layer.
quality_report.json	Machine-readable QA output.

Table 11: Primary project artefacts.

14. RECOMMENDATIONS AND FUTURE DEVELOPMENT

Future development will prioritise analytical robustness, validation and uncertainty handling.

- Run sensitivity analysis on HEIVA weighting and normalisation choices.
- Add population weighting where a policy question requires population impact rather than area ranking.
- Introduce boundary-version harmonisation for long historical comparisons.
- Expand the indicator set only when each new indicator has a clear decision-support purpose.
- Add confidence intervals / uncertainty communication to relevant dashboard views.
- Evaluate more appropriate time-series methods only if the source cadence and history justify them.
- Publish a lightweight data dictionary and model card and methodology note alongside the repository.
- Where permitted, deploy the Streamlit app and include the live URL in the GitHub README and CV portfolio section.

15. CONCLUSION

HEIVA England brings deprivation, population-health and economic-participation data into a single reproducible analytical workflow for English local authorities. The project combines automated ingestion and transformation, code-based geographic integration, relational storage, statistical analysis, composite scoring, historical analysis, clustering, geospatial processing and stakeholder-facing reporting.

The analysis identified a strong negative association between deprivation and healthy life expectancy ($r = -0.81$). The OLS model accounted for 66.5% of the observed cross-sectional variation in healthy life expectancy, with deprivation retaining a statistically significant negative association after economic inactivity was included. The HEIVA score further provides a transparent way to identify authorities where multiple area-level vulnerabilities coincide.

These results should be interpreted within the project's limitations. The analysis is ecological rather than individual-level, combines indicators with different reporting periods, and does not establish causal relationships. The HEIVA score is an experimental comparative measure rather than an official statistic or policy-allocation mechanism.

The resulting platform provides a reproducible basis for exploring health and socioeconomic inequality, while retaining sufficient transparency for its assumptions, data quality and analytical limitations to be examined and challenged.

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APPENDIX – REPOSITORY STRUCTURE

```
health-economic-inequality-england/  
├─ README.md  
├─ requirements.txt  
├─ config.py  
├─ main.py  
├─ app.py  
├─ data/  
│   ├─ raw/  
│   ├─ processed/  
│   ├─ geospatial/  
│   └─ database/  
├─ src/  
│   ├─ extract.py  
│   ├─ transform.py  
│   ├─ load_database.py  
│   ├─ analyse.py  
│   ├─ history.py  
│   ├─ forecasting.py  
│   ├─ clustering.py  
│   ├─ geospatial.py  
│   ├─ quality.py  
│   └─ load_advanced_database.py  
├─ dashboard/  
│   ├─ common.py  
│   ├─ overview.py  
│   ├─ trends.py  
│   ├─ segments.py  
│   ├─ explorer.py  
│   └─ methodology.py  
├─ sql/  
│   ├─ analysis_queries.sql  
│   └─ advanced_queries.sql  
├─ outputs/  
│   ├─ charts/  
│   ├─ tables/  
│   ├─ forecasts/  
│   └─ quality/  
├─ tests/  
│   └─ test_quality.py  
├─ .github/workflows/  
│   └─ refresh-heiva.yml  
└─ .streamlit/  
    └─ config.toml
```

GitHub Link: <https://github.com/Jirilin/health-economic-inequality-england.git>.